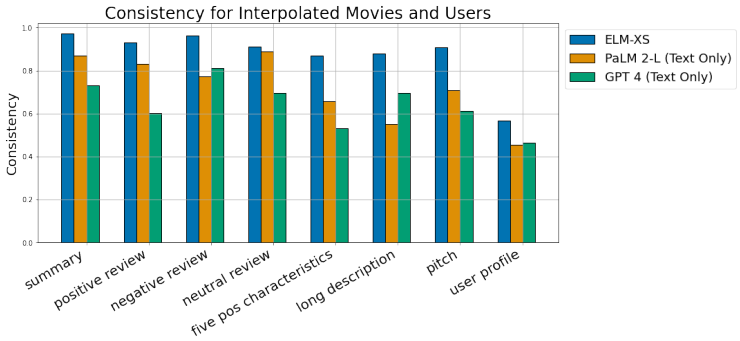

Figure 3: Results show consistency results for movie and user interpolations using ELM, compared to one-shot decoded results of text only LLMs. We show semantic consistency (cosine) scores for movies and behavioral consistency (NDCG) scores for user profiles.

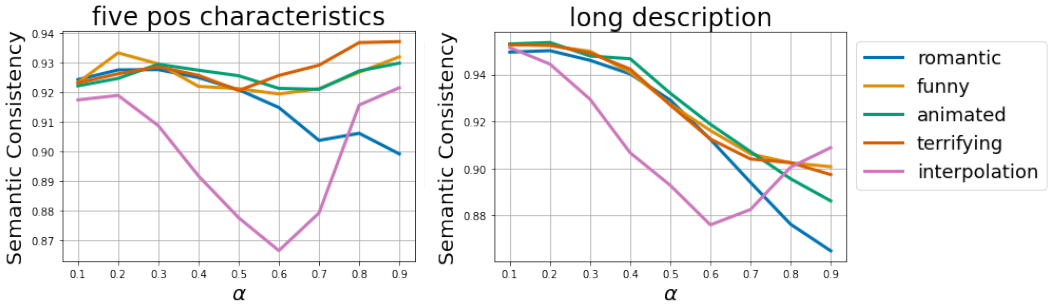

Figure 4: Plot depicts SC scores (using cosine similarity) for interpolations of movies with the movie "Forrest Gump", as well as extrapolation of CAV attributes in semantic space.

PaLM 2-XS (Otter). We use a two-layer MLP for the domain-embedding adapter layer E_A, and two-stage training, as described in Sec. 2. Particularly, for movie tasks, we run the first stage of training for 20k iterations, and then the second stage for another 300k iterations, using a batch size of 32 (i.e., roughly seven epochs). We use a similar procedure for user profiles. We found training in two stages to be essential for convergence of ELM (see Appendix D for further details).

Test-item Evaluation. Table 2 shows consistency (SC and BC) and human rater results for all movie tasks on the test set. For each task, we ask 100 human raters[8] to rate the consistency and comprehensiveness of our results w.r.t. the ground-truth data. We also ask qualitative task-specific questions (Appendix C describes rater instructions in detail). Rater evaluations show that ELM is able to generalize well across most tasks. Moreover ELM maintains high semantic consistency across all tasks and good behavioral consistency for most. We note that performance tends to be worse on two-movie tasks, due to the additional challenge of "conversing" about similarities and differences between two distinct embedding representations. We emphasize the importance of our consistency metrics, as they reflect how well ELM adapts to the domain embedding space $\mathcal{W}$.

Communicating Novel Entities. We next test ELM's capability to meaningfully extrapolate from existing movies or users to interpret "gaps" in embedding space (e.g., embedding vectors of hypothetical movies for which we predict a large audience). Note that such hypothetical vectors have no ground-truth data for evaluation; nor is it clear how to evaluate hypothetical embeddings using text-only LLMs (e.g., Google et al. (2023); OpenAI (2023)). For instance, there is no obvious way to instruct a text-only LLM to describe an arbitrary vector in $\mathcal{W}$. Even if we were to ask a text-only LLM to interpolate two existing entities (e.g., movies), the result will not necessarily be consistent with the domain embedding space we wish to interpret, nor can one in general direct a text-only LLM to an arbitrary point using interpolation between a small number of existing entities. Behavioral

[8]Raters were paid contractors. They received their standard contracted wage, which is above the living wage in their country of employment.

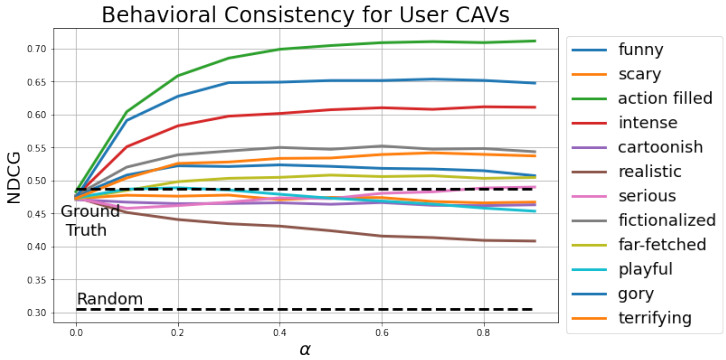

Figure 5: Plot depicts BC scores (using NDCG with the predicted CF model ratings) for user profile decoded outputs, where user embeddings are extrapolated in different CAV directions. Dotted lines show two baselines (for $\alpha = 0$): the random baseline, which rates movies randomly, and the ground-truth baseline, which uses the ground-truth user profile data to compute BC (i.e., BC of the ground-truth training data).

embeddings of, say, users are even more challenging for text-only LLMs. This puts additional emphasis on the need to evaluate such LLMs using embedding consistency metrics like SC and BC. That said, to ground our results, we do not consider arbitrary hypothetical points in embedding space. Instead, we evaluate our model using interpolation of *existing entities*, as well shifts of such entities in specific directions (see **Generalizing CAVs** below). Nevertheless, ELM's ability to interpret hypothetical embedding vectors is not constrained to these choices.

We first assess consistency metrics on interpolations of two movies or users. Fig. 3 compares the performance of ELM with state-of-the-art text-only LLMs; namely, PaLM 2-L and GPT 4. For movies, we linearly interpolate (with mixture coefficient $\alpha = 0.5$) embedding vectors of movies in the training set with the classic movie "Forrest Gump (1994)", and provide the *interpolated embedding* to ELM. For users, we interpolate random pairs of user embedding vectors from the training set. For a fairer comparison, we provide the text-only LLMs with a ground-truth examples of each task's output, and prompt them to generate the task output. Particularly for users, we provide the text-only LLMs with ground-truth user profiles, and instruct them to interpolate between them (see Appendix F for the precise prompts used). We measure semantic consistency for movie tasks and behavioral consistency for the user profile task. We found that text-only LLMs are significantly less consistent than ELM. This is not surprising, as describing a domain embedding space manifold to a text-only LLM is especially challenging. Moreover, ELM is able to generate high-quality, consistent responses to our tasks. We provide further qualitative results in Appendix G.

We next test ELM's consistency as we vary the degree of interpolation between movies, by varying the mixture weight α. Fig. 4 shows semantic consistency (magenta line) for two of our movie tasks. We find that outputs are highly consistent across all degrees of interpolation values, with a tendency for lower consistency scores for embeddings that are farther away from the two movies. This result is unsurprising, as we expect performance to degrade as it moves away from "real" data points. Also note these are linear interpolations, though the true embedding manifold may be highly non-linear.

Generalizing Concept Activation Vectors (CAVs). Finally, we test the interpretation of vectors in domain embedding space by extrapolating existing entity embeddings in specific directions. To accomplish this, we train CAVs [9] (Kim et al., 2018) using attribute tags in MovieLens 25M, provided by movie raters. These CAVs can be treated as noisy, "soft" movie attributes, such as 'funny,' 'terrifying,' 'thought-provoking,' etc. The CAV corresponding to a specific attribute provides a *direction* in domain embedding space that represents an increase in that attribute's value. We use these to *extrapolate* movies and users in specific CAV directions (e.g., make an existing movie funnier, or tweak a user profile to prefer more cartoonish movies).

In Fig. 4, we see that semantic consistency of movie extrapolations generally decreased as we extrapolated further in CAV directions. This is expected as extrapolations outside the data manifold should

[9]We train CAVs using the procedure of Göpfert et al. (2022) and the dataset of Balog et al. (2021), to which we refer for details. Specifically, we use linear, objective rather than non-linear and/or subjective CAVs.

induce higher error. Similarly, Fig. 5 shows behavioral consistency results for CAV extrapolations of user profiles. These results show attribute extrapolations to be behaviorally consistent for different α values. Perhaps surprisingly, user profiles show *greater* behavior consistency as we increase the degree to which they move in a CAV direction. We hypothesize this is due to user profiles becoming "more predictable" in our CF model by making user preferences more extreme. For instance, as we make a user profile gravitate to funnier movies, preferences tend to degenerate with a near-exclusive focus on funny movies (thus rendering rating prediction – and especially relative rating prediction – more accurate). This in turn increases BC. We provide further qualitative results for interpolations and CAV-based extrapolations in Fig. 1. We refer to Appendix G for more qualitative results.

5 Related Work

We review a selection of related work. The use of embeddings in NLP is well-established, with seminal work like *Word2Vec* (Mikolov et al., 2013) and *GloVe* (Pennington et al., 2014) using dense vectors to represent words in a way that captures semantics. These approaches serve as the foundation for many models in NLP, e.g., contextual (Bengio et al., 2000; Collobert & Weston, 2008; Radford et al., 2018) and deep-averaging networks (Iyyer et al., 2015). Bowman et al. (2015) show that interpolating sentences in embedding space can generate novel sentences that smoothly blend the "ideas" in the original sentences. Such NLP models provide some "immediate" interpretability.

Our ELM approach supports interpretation of *arbitrary* embedding spaces using natural language. In this sense, ELM can be seen as learning an encoder-decoder model (Raffel et al., 2020), for which the encoder adapts the domain-embedding to the token-embedding space, and the LLM acts as decoder from the "semantics" latent in the domain embedding to natrual language. As such, our work is closely related to research on translation (Vaswani et al., 2017) and multi-modality (Cho et al., 2021; Yu et al., 2022) that rely on encoder-decoder architectures. A key difference with ELM lies in the fact that encoder inputs (domain embeddings) are not igenerally interpretable; by contrast, in translation and multi-modal tasks, labeled data can be explicitly acquired from humans.

The use of continuous vectors as prompts for language models can provide flexibility in controlling outputs, improving performance on downstream tasks. Examples include: prefix tuning (Li & Liang, 2021; Lester et al., 2021; Tsimpoukelli et al., 2021), which prepends a sequence of task-specific vectors to the input, while keeping LLM parameters frozen; and hard-soft prompt hybrid tuning (Liu et al., 2021; Han et al., 2022), which inserts tunable embeddings into a hard prompt template. These methods primarily guide model behavior for downstream tasks, rather than *interpreting* embeddings, especially those derived from different data sources (which we want to understand, not create).

CAVs (Kim et al., 2018) and visual techniques such as t-SNE (Van der Maaten & Hinton, 2008) or UMAP (McInnes et al., 2018) have been used for visualizing and interpreting embeddings. Nevertheless, these methods often provide static insights or visual interpretations, and are often hard or even impossible to capture in a low dimension (Chari & Pachter, 2023; Seshadhri et al., 2020). ELM, by contrast, seeks to enable dynamic, language-based exploration of the embedding space, "visualizing" it by narrating embeddings with language. Other work has focused on interpretability analysis and toolkits for language and sequence models (Belinkov et al., 2020; Sarti et al., 2023; Bills et al., 2023; Räuker et al., 2023), but these methods are largely orthogonal to our work.

6 Conclusion

We have presented ELM, a novel language-model framework for interpreting domain-embedding spaces. We assessed ELM's capabilities on a range of movie tasks and a user-profile task, benchmarking with both human evaluations as well as two novel quantitative metrics; namely, semantic and behavioral consistency. Our results show that ELM generalizes well to embedding vectors in a test set, and aligns well with human-rater expectations. We also find that ELM is adept at handling the nuanced challenge of describing novel entities, with better semantic and behavioral in tasks that interpolate between entities, compared to state-of-the-art text-only LLMs. We also demonstrated ELM's proficiency in generalizing CAVs, underscoring its ability to meaningfully manipulate and interpret domain attributes of movies and user profiles. Taken together, our results suggest that ELM offers a powerful, flexible mechanism for understanding, navigating and manipulating complex embedding representations.